

DICOM Q/R Pull Agent — Installation Guide

DiagnoCenter ERP | May 2026 |
Technician Copy

Automatic study retrieval: imaging machines (MRI / CT / X-Ray / USG) → Conquest PACS → ERP Radiology Worklist

What this does

After completing these 8 steps, Conquest PACS will automatically pull new studies from every imaging machine at a configured interval, with no technician action required. Studies appear in the ERP Radiology Worklist within seconds of being pulled into Conquest.

```
Patient scanned on machine  
-> Agent runs findscu (C-  
-> Agent runs movescu (C-  
-> ERP Worklist updated a  
-> Radiologist opens study
```

Pre-flight checklist — confirm before starting

- Conquest PACS is installed and running on this PC
- This PC is on the same LAN as all imaging machines
- You have the ERP URL (e.g. <https://yourapp.replit.app>)
- You have the INTERNAL_API_KEY value (from ERP Admin → Secrets)
- You know each machine's IP address and DICOM AE Title

1 Install Node.js on the Conquest PC

Node.js is the runtime that the pull agent uses. It is free and safe to install alongside Conquest.

Download: <https://nodejs.org> → choose the **LTS** version → run the installer with all default options.

After installation, open **Command Prompt** (search "cmd" in the Start menu) and verify:

```
node --version
```

Expected output (version number may differ):

```
v22.4.0
```

Note: If "node" is not recognised after installing, restart the Command Prompt or reboot the PC. The PATH update requires a fresh shell session.

2 Install DCMTK (DICOM command-line tools)

DCMTK provides **findscu** (queries machines for studies) and **movescu** (pulls them into Conquest). These are free, industry-standard DICOM tools used by radiology departments worldwide.

Download: <https://dcmkt.org> → Downloads → DCMTK → **Win64** ZIP (latest stable release).

Extract the ZIP to a permanent folder on this PC:

```
C:\DCMTK\
```

Inside you will find a **bin** sub-folder containing findscu.exe, movescu.exe, echoscu.exe and others.

Verify the installation in Command Prompt:

```
C:\DCMTK\bin\findscu --version
```

Expected output starts with:

```
$dcmkt: findscu v3.6.x ...
```

Note: Do NOT add DCMTK to the system PATH. The agent locates it via the DCMTK_DIR environment variable you set in Step 4.

3 Copy the Pull Agent files to the Conquest PC

From the ERP project folder, copy the entire **dicom-pull-agent** directory to this PC. Place it at:

```
C:\DicomPullAgent\
```

The final folder structure should look exactly like this:

```
C:\DicomPullAgent\  
  src\  
    index.js      <- the agent script (main file)  
    package.json  
    README.md
```

Result: The agent has zero npm dependencies. No need to run npm install. Just copy the files and it is ready to run immediately.

Copy options: USB drive, shared network folder, or download directly from the ERP project.

4 Create the startup batch file

Open **Notepad**, paste the content below, fill in your actual values, then save as:

```
C:\DicomPullAgent\start-agent.bat
```

```

@echo off
REM -- DiagnoCenter DICOM Pull Agent Startup -----
REM Edit the values below, then save this file.

REM Your published ERP URL
set ERP_BASE_URL=https://YOUR-APP-NAME.replit.app

REM Must match INTERNAL_API_KEY in ERP environment secrets
set INTERNAL_API_KEY=PASTE-YOUR-KEY-HERE

REM AE Title this agent presents when connecting to modalities
set AGENT_AE_TITLE=DIAGNO_AGENT

REM Conquest PACS settings (check dicom.ini: MyACRNema and TCPPort)
set CONQUEST_AE_TITLE=CONQUEST1
set CONQUEST_HOST=127.0.0.1
set CONQUEST_PORT=5678

REM Path to DCMTK bin\ folder (where findscu.exe lives)
set DCMTK_DIR=C:\DCMTK\bin

REM Poll interval in milliseconds (30000 = 30 seconds)
set POLL_INTERVAL_MS=30000

REM Logging: debug | info | warn | error
set LOG_LEVEL=info

REM -- Start the agent -----
node C:\DicomPullAgent\src\index.js
pause

```

Where to find each value:

Variable	Where to find the value
ERP_BASE_URL	Your ERP published URL (e.g. https://diagnocenter.replit.app)
INTERNAL_API_KEY	ERP Admin -> Environment Secrets -> INTERNAL_API_KEY
CONQUEST_AE_TITLE	Open dicom.ini -> find line: MyACRNema=
CONQUEST_PORT	Open dicom.ini -> find line: TCPPort=
DCMTK_DIR	The bin\ folder inside your DCMTK extraction directory

5 Register machines in Conquest's dicom.ini

Conquest needs to know each imaging machine's AE Title and IP address. Open **dicom.ini** (in the Conquest installation folder, e.g. C:\Conquest\)) and find the **[ACRNema]** section. Add one line per device:

```
[ACRNema]
; Format: AETitle = IP-Address Port
MRI_SIEMENS = 192.168.1.10 104
CT_GE = 192.168.1.11 104
XRAY_FUJI = 192.168.1.12 104
USG_MINDRAY = 192.168.1.13 104

; Register the pull agent so modalities accept C-MOVE requests from it
DIAGNO_AGENT = 127.0.0.1 9999
```

Note: Use the EXACT AE Titles configured on each machine (case-sensitive). To find a machine's AE Title: go into its DICOM settings menu and look for 'AE Title' or 'SCU AE Title'. Also note the machine's IP address and DICOM port (usually 104 or 11112).

Save dicom.ini and restart Conquest for changes to take effect.

6 Test the agent manually

Double-click **C:\DicomPullAgent\start-agent.bat**. A Command Prompt window opens and you should see:

```
+-----+
| DiagnoCenter DICOM Q/R Pull Agent |
+-----+
ERP URL   : https://your-erp.replit.app
Agent AE  : DIAGNO_AGENT
Conquest  : CONQUEST1@127.0.0.1:5678
Poll every : 30s
Agent ID   : CONQUEST-PC-NAME

DCMTK findscu found v
[agent] Running -- polling every 30s. Press Ctrl+C to stop.
```

Test a manual pull from the ERP:

1. Log in to ERP → open **DICOM Nodes**
2. Find any machine → click the **Pull Now** button (download icon)
3. Switch to the **Pull Jobs** tab — watch the status change: pending → running → completed
4. Check the Conquest viewer — the pulled studies should appear

Note: If the job shows 'failed': the agent Command Prompt window prints the DCMTK error. Common causes: wrong machine IP, AE Title mismatch, machine Q/R SCP not enabled, or Windows Firewall blocking port 104. Check each machine's DICOM service settings to ensure Query/Retrieve SCP is enabled.

7 Install as a Windows Service (auto-start with Windows)

Use **NSSM** (Non-Sucking Service Manager) — a free, single-file tool that runs any program as a Windows Service. No programming required.

Download NSSM: <https://nssm.cc/download> → extract → copy nssm.exe to C:\DicomPullAgent\nssm.exe

Open Command Prompt **as Administrator** (right-click → Run as administrator):

```
C:\DicomPullAgent\nssm.exe install DicomPullAgent
```

A GUI window opens. Fill in each tab as follows:

Tab	Field	Value to enter
Application	Path	C:\Program Files\nodejs\node.exe
Application	Startup directory	C:\DicomPullAgent
Application	Arguments	src\index.js
Environment	(one line each)	All variables from Step 4 without the 'set ' prefix
Details	Display name	DiagnoCenter DICOM Pull Agent

Click **Install service**. Then start it:

```
C:\DicomPullAgent\nssm.exe start DicomPullAgent
```

Verify it is running:

```
sc query DicomPullAgent

(Expected output includes:  STATE : 4  RUNNING)
```

Result: The agent now starts automatically every time Windows boots, even without anyone logged in -- exactly like Conquest itself.

8 Enable Auto-Pull in the ERP for each machine

The final step is to turn on the automatic schedule in the ERP for each imaging machine:

1. Log in to ERP → go to **DICOM Nodes**
2. Click **Edit** on the first machine (e.g. CT Scanner)
3. Scroll to **Auto-Pull (DICOM Q/R)** → toggle it **ON**
4. Set **Pull frequency** — recommended: *Every 15 minutes* for active machines
5. Set **Query days back** — *Today only* is sufficient for live operation
6. Leave Conquest fields blank (agent uses defaults from Step 4)
7. Click **Save Changes**
8. Repeat for MRI, X-Ray, and USG machines

Result: What happens next: The ERP cron runs every 5 minutes. When a machine's interval is due, it creates a pull job. The agent on this PC picks it up within 30 seconds, queries the machine for today's studies, and pulls each one into Conquest automatically. Studies appear in the ERP Radiology Worklist with no technician action required.

Quick Reference — Common Commands (run as Administrator in CMD)

Task	Command
Start service	nssm start DicomPullAgent
Stop service	nssm stop DicomPullAgent
Restart service	nssm restart DicomPullAgent
View service status	sc query DicomPullAgent
Edit service settings	nssm edit DicomPullAgent
Remove service	nssm remove DicomPullAgent confirm
Test DCMTK version	C:\DCMTK\bin\findscu --version
Test Conquest echo	C:\DCMTK\bin\echoscu -aet DIAGNO_AGENT -aec CONQUEST1 127.0.0.1 5678